Junyan Liu

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Education

Columbia University in the City of New York — New York, US

MS in Mechanical Engineering (Robotics and Control Track) — *Sep 2025 – Current*

- Coursework: Robotic Studio (Advanced), Robot Learning, Model Predictive Control, Introduction to Robotics

University of Wollongong — Wollongong, AU

BA in Mechatronic Engineering (Honours) — *Jul 2024 – Jun 2025*

- Coursework: Applied Topics in Mechatronics, Reliability Engineering, Manufacturing Engineering Principles

Beijing Jiaotong University — Beijing, CN

BA in Mechatronic Engineering — *Sep 2021 – Jun 2025*

- Coursework: Electronics, Programming for Engineers, Embedded System and Applications

Honors and Awards

Winner, Best Use of Computer Vision — MakeCU 2025 Hackathon (Columbia Robotics Club) — Nov 2025

Research Experience

World Action Model for Robotic Manipulation — Jun 2026 – Present

RoboPIL Lab, Prof. Yunzhu Li (mentored by PhD student Kaifeng Zhang)

- Explored training-free control baselines for manipulation, including MPPI-based grasping with a Franka arm in the Newton simulator and MPPI-based atomic-task planning in RoboCasa, before converging on a learning-based direction.
- Currently developing a lightweight World Action Model (WAM) by extending the video-generation world model from Interactive World Simulator with a trainable action head, aiming to produce a baseline policy for manipulation evaluation.
- Iterating on the action-head architecture and training pipeline (in progress).

Replicate Project: Self-Reproducing Modular Robots — Feb 2026 – Present

Creative Machines Lab, Prof. Hod Lipson

- Co-developed the teleoperated control system enabling physical module collection and docking: a browser-based control terminal issues discrete motion commands over a serial link to a host microcontroller, which wirelessly broadcasts commands to modules and receives periodic heartbeat status for real-time operator guidance.
- Programmed microcontroller (Arduino Nano ESP32) firmware implementing discrete motion primitives (forward step, left/right turn) and a pre-programmed separation sequence, enabling a composite parent-offspring structure to reliably split into two independent, fully functional robots.
- Engineered a low-level magnetic physics engine to overcome MuJoCo's lack of native complex magnetic support, formulating an $O(N^2)$ magnetic calculation model with non-linear distance attenuation and velocity damping that validated the docking-force parameters later adopted for the physical magnetic docking interface.
- Spearheaded the strategic migration of the simulation framework to DeepMind's MuJoCo/MJX, configuring high-performance cloud servers (vCPU) and local Windows GPU acceleration to optimize large-scale evaluations for random search and genetic algorithms.
- Overcame Python's matrix processing bottlenecks by developing a highly optimized C dynamic library, bypassing the Python interpreter to directly inject force data into MuJoCo's low-level array, achieving a 2.34x speedup per simulation step.

Influence of Magnet Internal Structure on Magnetic Performance — Aug 2024 – Jun 2025

University of Wollongong, Dr. Chin-Hsing Kuo

- Investigated how internal hollow structures affect magnetic force in cylindrical and rectangular NdFeB magnets via dual-platform simulation: MATLAB (Monte Carlo dipole-dipole modeling with 1,000,000+ sampling points per configuration) and ANSYS Maxwell 2024 (finite element analysis using the Virtual Work method).
- Developed a custom MATLAB simulation pipeline (dipole-dipole interaction model, systematic geometry sweeps, cubic polynomial curve fitting) to extract force-geometry relationships, cross-validated against ANSYS Maxwell FEM results.
- Identified a non-linear, cubic relationship between internal hollow size and magnetic force; found rectangular magnets exhibit higher initial force but a steeper decline than cylindrical magnets, attributable to edge and corner flux-distortion effects.
- Presented research findings to academic supervisors and peers, communicating technical results to both engineering and non-technical audiences.

Legged Robot Learns to Walk via Reinforcement Learning Jan 2024 – Feb 2024

Global Education, Academics, and Research Skills (GEARS)

- Configured a Ubuntu-based RL simulation environment and designed reward functions to train a legged robot controller, evaluating walking performance across different reward compositions.

Publications

S. Zhang*, **J. Liu***, W. He*, H. Lipson, "One Module Design, Two Morphologies: Toward Self-Reproducing Modular Robots," *submitted to IROS 2026 Workshop*, 2026 (under review).

*Equal contribution.

Teaching Experience

Columbia University New York, US

Teaching Assistant *Sep 2026 – Present*

MECE4058E: Mechatronics & Embedded Micro *Instructor: Enrico Zordan*

- Helped students understand and complete lab exercises in embedded microcontroller programming (Assembly/C) and mechatronic systems (solenoids, stepper/DC motors, thermal systems, magnetic levitation), and graded lab results.

Internship Experience

C-Based Structural Simulation Software Developer Nov 2024 – Feb 2025

Shenzhen Yunbo Software Technology Co., Ltd.

- Developed and optimized structural simulation software in C, enhancing performance of model setup, simulation core, and result visualization modules; improved simulation efficiency by 20%.
- Collaborated with a cross-functional team of 7 engineers to identify requirements and optimize algorithms, ensuring accuracy within 2% deviation from benchmark data.
- Conducted 50+ test/debug cycles, reducing runtime errors by 30% and increasing software stability.

Projects Design

Robotic Studio: Quadruped Robot "Pack-Pack" Development Sep 2026 – Dec 2026

- Designed and developed a custom quadruped robot from concept sketches to physical deployment, completing the entire pipeline of mechanical design, simulation, and control.
- Engineered the mechanical structure using SolidWorks for 3D CAD modeling and assembled the physical prototype integrating a Raspberry Pi core controller and 8 servo motors (2 DOF per leg).
- Constructed a high-fidelity physics simulation environment utilizing MuJoCo to safely and efficiently design, test, and evaluate various locomotion control strategies.
- Implemented and evaluated multiple control algorithms within the simulation, including preset gaits, Random Search, Hill Climbing, and Reinforcement Learning (RL), analyzing models optimized for maximum speed and gait stability.
- Successfully achieved sim-to-real transfer of heuristic search algorithms, deploying locomotion policies optimized via Random Search and Hill Climbing onto the physical hardware to achieve a stable real-world walking speed of 32.33 cm/s.

Google It: Computer-Vision Tracking Robot (MakeCU 2025 Hackathon) Nov 2025

- Built an autonomous mobile robot with a 4-person team in a 24-hour hackathon; pivoted from an initial two-wheel self-balancing design (PID control via MPU6050 IMU) to a three-wheel, camera-guided tracking robot after balance tuning proved infeasible within the time limit.
- Implemented real-time ArUco marker detection and tracking on a Raspberry Pi using OpenCV, computing proportional steering corrections from the marker's pixel offset to drive LX-16A bus-servo-actuated wheels toward the target.
- Integrated an HC-SR04 ultrasonic sensor for obstacle avoidance and an LED status indicator, halting the robot when an obstacle was detected or the target was lost.
- Diagnosed and fixed hardware failures under time pressure — burned-out motors, slipping wheels (resolved with paper shims and wire-wrapped tires), and insufficient battery current (upgraded from 9V to 12V with a DC-DC converter).

Lumbar Surgery Intelligent Planning System May 2023 – May 2024

- Developed an intelligent planning system for lumbar surgery based on X-ray images.
- Utilized deep learning algorithms (U-Net) for automatic lumbar parameter measurements and combined previous surgical data for intelligent surgery planning.
- Aimed to improve surgery efficiency and accuracy, reducing risks through intelligent decision-making support; collaborated with a team under the supervision of Prof. Tian Lixia.

Mechatronics Engineering Cup – Smart Car Challenge Mar 2023 – Jul 2023

- Built an STM32F103VCT-based smart car integrating infrared, ultrasonic, flame, photoresistor, and color sensors to achieve obstacle avoidance, line tracking, fire extinguishing, color-based sorting, and light-source-seeking behaviors.
- Implemented the embedded control stack: PWM motor control, ADC/DAC signal processing, SPI/I2C sensor interfacing, USART communication, and OLED display output.
- Designed individual wheel control logic and integrated it into unified car operation; tested and optimized across multiple terrains to validate reliable multi-function performance.

Ice Mark Grip Oct 2021 – Dec 2021

- Designed a procedure to command the robot's movement in two dimensions and search for objects using sensors.
- Applied hinges, sensors, and basic parts to allow motors to control planes of motion; the robot approaches with water after receiving a signal from the switch, and dispenses water after sensing the object's weight.
- Sensed instructions from the switch: the robot carries water toward the object from a distance and delivers it after serving.

Technical Skills

Programming & Embedded Systems: C/C++, Python, MATLAB, Arduino (ESP32), STM32
Robotics, Simulation & ML: MuJoCo/MJX, Deep Learning (U-Net)
Design & Analysis: SOLIDWORKS, AutoCAD, Inventor, Photoshop, ANSYS Workbench, ANSYS Maxwell
Tools & OS: Linux/Ubuntu, Git